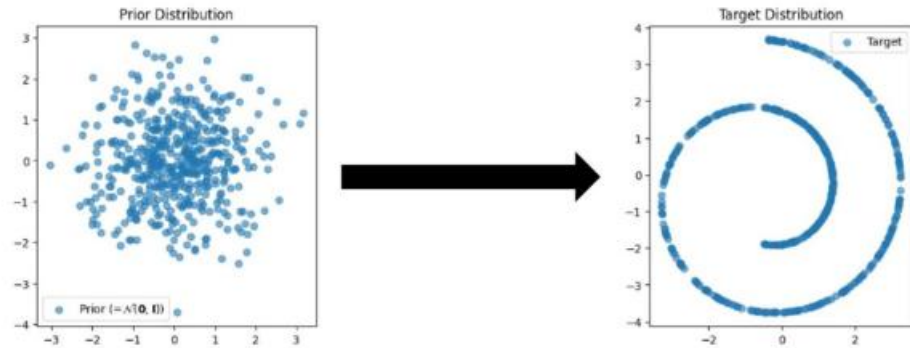


Lab-1 DDPM



Task 1 - Swiss Roll



Task 2 - Image Generation

TA: Yi-Ruei Liu

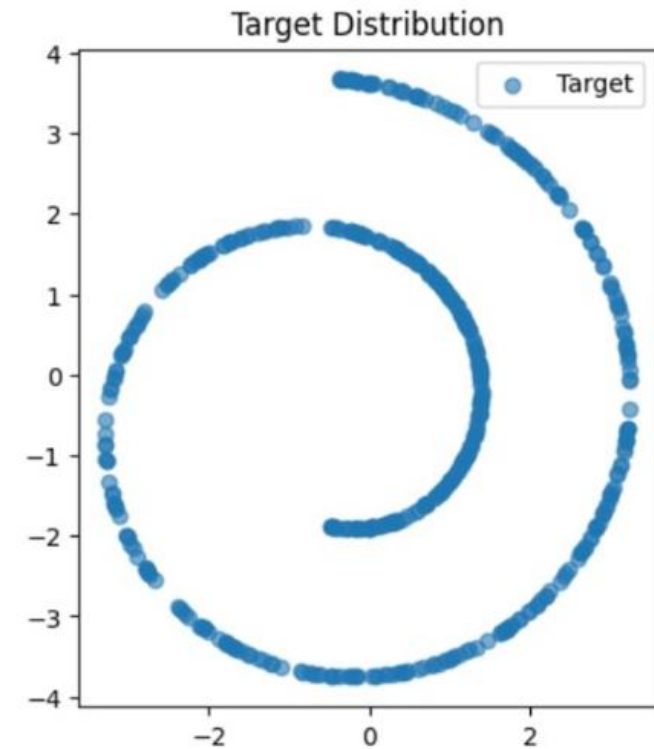
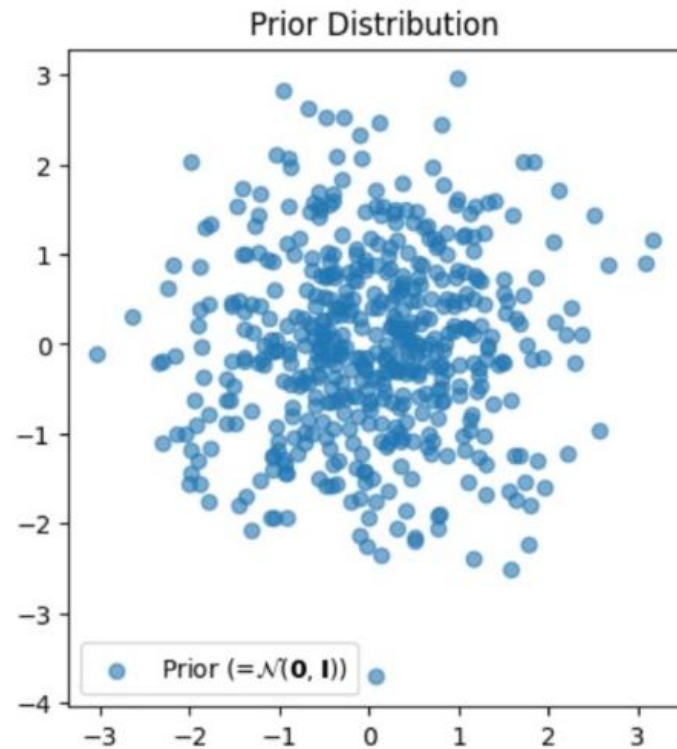
Due: 2026/10/1 (Thu.) 23:59

Links

- [DDPM paper](#)
- [Starter code](#)

Task1 – Swiss Roll

We're going to model a simple “Swiss Roll” 2D distribution.



Task1 – Swiss Roll

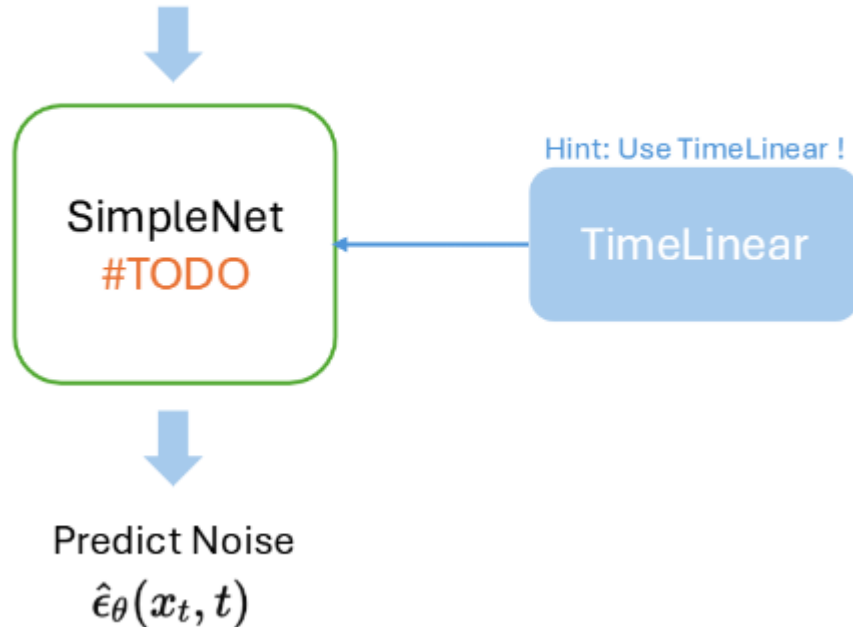
- You'll implement key components of training a DDPM:
 - SimpleNet: prediction network
 - q_sample: forward process
 - p_sample: reverse process
 - compute loss

Task1 – Swiss Roll

TODO #1 – SimpleNet

- 2d_plot_diffusion_todo/network.py

1. Noise data x_t
2. Current timestep t



```
class SimpleNet(nn.Module):
    def __init__(
        self, dim_in: int, dim_out: int, dim_hids: List[int], num_timesteps: int
    ):
        super().__init__()
        """
        (TODO) Build a noise estimating network.

        Args:
            dim_in: dimension of input
            dim_out: dimension of output
            dim_hids: dimensions of hidden features
            num_timesteps: number of timesteps
        """

        ##### TODO #####
        # DO NOT change the code outside this part.
        #####

    def forward(self, x: torch.Tensor, t: torch.Tensor):
        """
        (TODO) Implement the forward pass. This should output
        the noise prediction of the noisy input x at timestep t.

        Args:
            x: the noisy data after t period diffusion
            t: the time that the forward diffusion has been running
        """

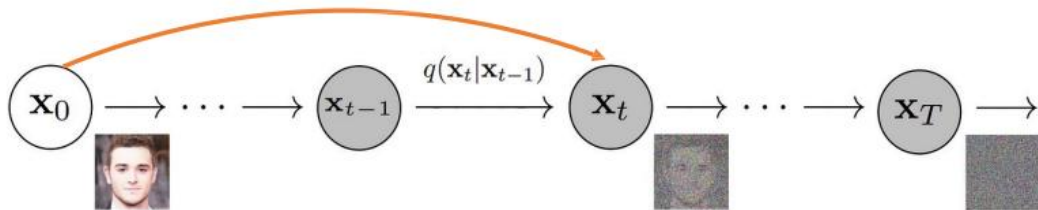
        ##### TODO #####
        # DO NOT change the code outside this part.
        #####

        return x
```

Task1 – Swiss Roll

TODO #2 – q_sample

- 2d_plot_diffusion_todo/ddpm.py



$$q(x_t | x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

$$\text{where } \bar{\alpha}_t = \prod_{i=1}^t \alpha_i.$$

```
def q_sample(self, x0, t, noise=None):
    """
    sample x_t from q(x_t | x_0) of DDPM.

    Input:
        x0 (`torch.Tensor`): clean data to be mapped to timestep t in the forward process of DDPM.
        t (`torch.Tensor`): timestep
        noise (`torch.Tensor`, optional): random Gaussian noise. if None, randomly sample Gaussian noise i
    Output:
        xt (`torch.Tensor`): noisy samples
    """
    if noise is None:
        noise = torch.randn_like(x0)

    ##### TODO #####
    # DO NOT change the code outside this part.
    # Compute xt.
    alphas_prod_t = extract(self.var_scheduler.alphas_cumprod, t, x0)
    xt = x0

    #####

    return xt
```

Task1 – Swiss Roll

TODO #2 – q_sample

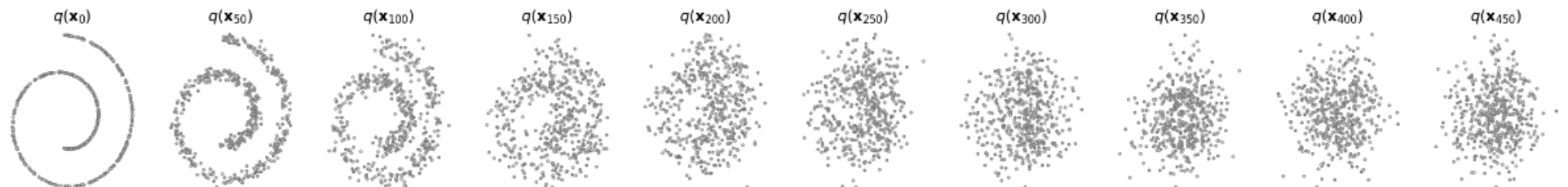
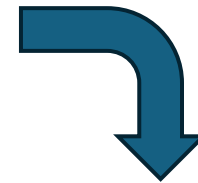
- 2d_plot_diffusion_todo/ddpm_tutorial.ipynb

Visualize $q(\mathbf{x}_t)$

```
fig, axes = plt.subplots(1, 10, figsize=(28, 3))
for i, t in enumerate(range(0, 500, 50)):
    x_t = ddpm.q_sample(target_ds[:num_vis_particles].to(device), (torch.ones(num_vis_par
    x_t = x_t.cpu()
    axes[i].scatter(x_t[:,0], x_t[:,1], color='white', edgecolor='gray', s=5)
    axes[i].set_axis_off()
    axes[i].set_title('$q(\mathbf{x}_{'+str(t)+'})$')
```

Python

You'll see this if you correctly
implement `q_sample()`



Task1 – Swiss Roll

TODO #3 – p_sample

- 2d_plot_diffusion_todo/ddpm.py

How to get $x_t \xrightarrow{p_\theta} x_{t-1}$?

1. Predict noise from SimpleNet

$$2. \mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \hat{\epsilon}_\theta(x_t, t) \right) \quad \text{Eq. 11}$$

$$3. \tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t} \beta_t \quad \text{Sec. 3.2}$$

$$4. x_{t-1} = \mu_\theta(x_t, t) + \sqrt{\tilde{\beta}_t} z, \quad z \sim \mathcal{N}(0, I) \quad \text{Eq. 11}$$

```
def p_sample(self, xt, t):
    """
    One step denoising function of DDPM:  $x_t \rightarrow x_{t-1}$ .
    Input:
        xt ('torch.Tensor'): samples at arbitrary timestep t.
        t ('torch.Tensor'): current timestep in a reverse process.
    Output:
        x_t_prev ('torch.Tensor'): one step denoised sample. (=  $x_{t-1}$ )
    """
    ##### TODO #####
    # DO NOT change the code outside this part.
    # compute x_t_prev.
    if isinstance(t, int):
        t = torch.tensor([t]).to(self.device)
    eps_factor = (1 - extract(self.var_scheduler.alphas, t, xt)) / (
        1 - extract(self.var_scheduler.alphas_cumprod, t, xt)
    ).sqrt()

    beta_t = extract(self.var_scheduler.betas, t, xt) #  $\beta_t$ 
    alpha_t = extract(self.var_scheduler.alphas, t, xt) #  $\alpha_t$ 
    alpha_bar_t = extract(self.var_scheduler.alphas_cumprod, t, xt) #  $\bar{\alpha}_t$ 
    t_prev = (t - 1).clamp(min=0)
    alpha_bar_t_prev = extract(self.var_scheduler.alphas_cumprod, t_prev, xt) #  $\bar{\alpha}_{t-1}$ 

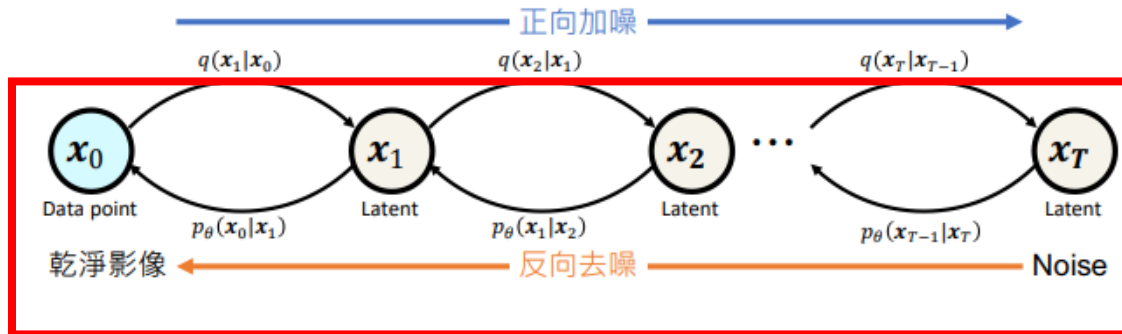
    # 1. predict noise
    # 2. Posterior mean
    # 3. Posterior variance
    # 4. Reverse step

    #####
    return x_t_prev
```

Task1 – Swiss Roll

TODO #4 – p_sample_loop

- 2d_plot_diffusion_todo/ddpm.py



```
def p_sample_loop(self, shape):
    """
    The loop of the reverse process of DDPM.

    Input:
    |   shape (`Tuple`): The shape of output. e.g., (num particles, 2)
    Output:
    |   x0_pred (`torch.Tensor`): The final denoised output through the DDPM reverse process.
    """
    ##### TODO #####
    # DO NOT change the code outside this part.
    # sample x0 based on Algorithm 2 of DDPM paper.
    x0_pred = torch.zeros(shape).to(self.device)

    #####
    return x0_pred
```

Task1 – Swiss Roll

TODO #5 – compute loss

- 2d_plot_diffusion_todo/ddpm.py

$$\mathbb{E}_{t>1, q(x_t|x_0)} [\|\hat{\epsilon}_\theta(x_t, t) - \epsilon_t\|^2]$$

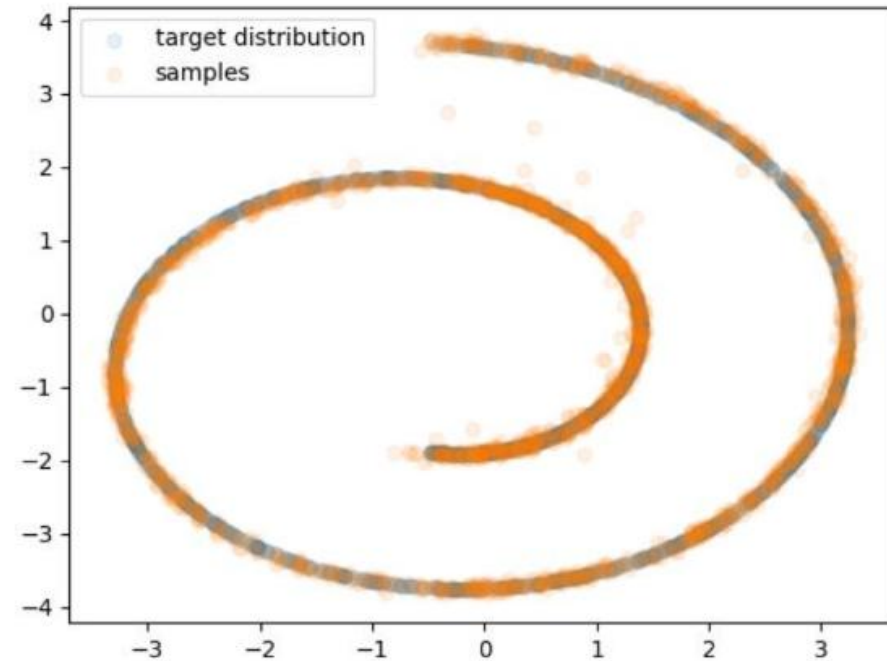
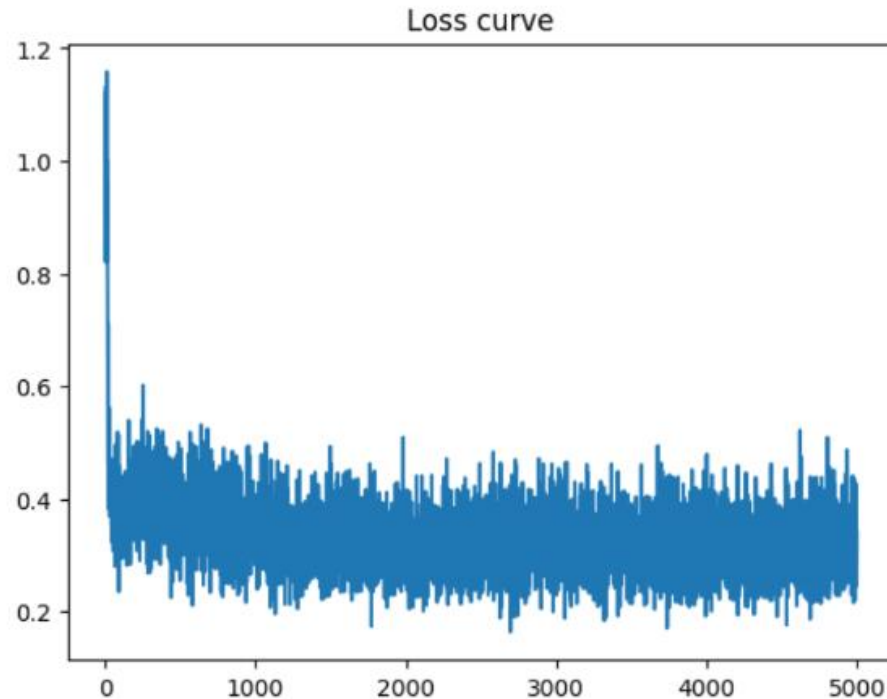
```
def compute_loss(self, x0):  
    """  
    The simplified noise matching loss corresponding Equation 14 in DDPM paper.  
    Input:  
    |   x0 (`torch.Tensor`): clean data  
    Output:  
    |   loss: the computed loss to be backpropagated.  
    """  
  
    ##### TODO #####  
    # DO NOT change the code outside this part.  
    # compute noise matching loss.  
    batch_size = x0.shape[0]  
  
    # 1) random choose timestep  
    t = (  
        torch.randint(0, self.var_scheduler.num_train_timesteps, size=(batch_size,))  
        .to(x0.device)  
        .long()  
    )  
    # 2) get GT noise, and use q_sample to get x_t  
  
    # 3) predict noise  
  
    # 4) MSE loss (eps, eps_pred)  
  
    loss = None  
  
    #####  
    return loss
```

Task1 – Swiss Roll

TODO #6 – train & evaluate

- 2d_plot_diffusion_todo/ddpm_tutorial.ipynb

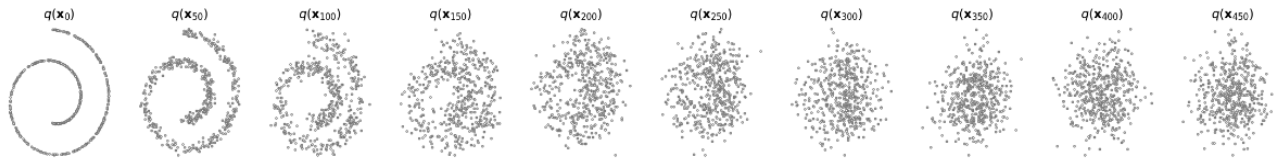
You'll see this if you correctly
Implement TODO #1 - #5



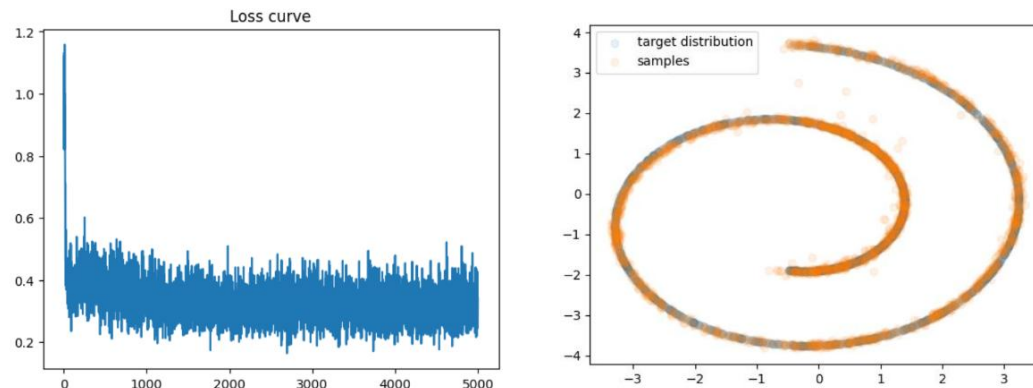
Task1 – Swiss Roll

TODO #7 – report

- Complete and explain the implementation of all TODO codes (20pts)
- Figure of q_{sample} implementation (10pts)



- Figure of the loss curve and the output distribution (10pts)



Task2 – Image Generation

Let's model a higher dimension distribution: Images!



Task2 – Image Generation

- You'll implement the similar things in task 1, and
 - Different beta scheduler: linear, quadratic, and cosine
 - Different prediction target: noise, x_0 , mean

Task2 – Image Generation

TODO #1 – add_noise

- image_diffusion_todo/scheduler.py

Hint: similar to q_sample

```
def add_noise(
    self,
    x_0: torch.Tensor,
    t: torch.IntTensor,
    eps: Optional[torch.Tensor] = None,
):
    """
    A forward pass of a Markov chain, i.e.,  $q(x_t | x_0)$ .

    Input:
    | x_0 ('torch.Tensor [B,C,H,W]'): samples from a real data distribution  $q(x_0)$ .
    | t: ('torch.IntTensor [B]')
    | eps: ('torch.Tensor [B,C,H,W]', optional): if None, randomly sample Gaussian noise in the function.
    Output:
    | x_t: ('torch.Tensor [B,C,H,W]'): noisy samples at timestep t.
    | eps: ('torch.Tensor [B,C,H,W]'): injected noise.
    """

    if eps is None:
        eps = torch.randn(x_0.shape, device='cuda')

    ##### TODO #####
    # DO NOT change the code outside this part.
    # Assignment 1. Implement the DDPM forward step.
    x_t = None

    #####

    return x_t, eps
```

Task2 – Image Generation

TODO #2 – beta_scheduler

- image_diffusion_todo/scheduler.py

$$q(x_t | x_0) = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

$$\alpha_t = 1 - \beta_t$$

$$\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$$

$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, f(t) = \cos\left(\frac{t/T + s}{1 + s} \cdot \frac{\pi}{2}\right)^2$$

```
if mode == "linear":
    betas = torch.linspace(beta_1, beta_T, steps=num_train_timesteps)
elif mode == "quad":
    betas = (
        torch.linspace(beta_1**0.5, beta_T**0.5, num_train_timesteps) ** 2
    )
elif mode == "cosine":
    ##### TODO #####
    # Implement the cosine beta schedule (Nichol & Dhariwal, 2021).
    # Hint:
    # 1. Define alphā_t = f(t/T) where f is a cosine schedule:
    #     alphā_t = cos^2( ( (t/T + s) / (1+s) ) * (π/2) )
    #     with s = 0.008 (a small constant for stability).
    # 2. Convert alphā_t into betas using:
    #     beta_t = 1 - alphā_t / alphā_{t-1}
    # 3. Return betas as a tensor of shape [num_train_timesteps].
    raise NotImplementedError("TODO: Implement cosine beta schedule here!")
```

Task2 – Image Generation

TODO #3 – predictor

- image_diffusion_todo/scheduler.py

Implement three types of predictors:

Mean predictor $\mu_{\theta}(x_t, t)$

x_0 predictor $\hat{x}_{\theta}(x_t, t)$

ϵ_t predictor $\hat{\epsilon}_{\theta}(x_t, t)$

Hint: recall what you've done in p_sample

```
def step_predict_noise(self, x_t: torch.Tensor, t: int, eps_theta: torch.Tensor):
    """
    Noise prediction version (the standard DDPM formulation).

    Input:
        x_t: noisy image at timestep t
        t: current timestep
        eps_theta: predicted noise  $\hat{\epsilon}_{\theta}(x_t, t)$ 
    Output:
        sample_prev: denoised image sample at timestep t-1
    """
    ##### TODO #####
    # 1. Extract beta_t, alpha_t, and alpha_bar_t from the scheduler.
    # 2. Compute the predicted mean  $\mu_{\theta}(x_t, t) = 1/\sqrt{\alpha_t} * (x_t - (\beta_t/\sqrt{1-\alpha_t})) * \dots$ 
    # 3. Compute the posterior variance  $\tilde{\beta}_t = ((1-\alpha_{t-1})/(1-\alpha_t)) * \beta_t$ .
    # 4. Add Gaussian noise scaled by  $\sqrt{\tilde{\beta}_t}$  unless t == 0.
    # 5. Return the final sample at t-1.
    sample_prev = None
    #####
    return sample_prev
```

```
def step_predict_x0(self, x_t: torch.Tensor, t: int, x0_pred: torch.Tensor):
    """
    x0 prediction version (alternative DDPM objective).

    Input:
        x_t: noisy image at timestep t
        t: current timestep
        x0_pred: predicted clean image  $\hat{x}_{\theta}(x_t, t)$ 
    Output:
        sample_prev: denoised image sample at timestep t-1
    """
    ##### TODO #####

    sample_prev = None
    #####
    return sample_prev

def step_predict_mean(self, x_t: torch.Tensor, t: int, mean_theta: torch.Tensor):
    """
    Mean prediction version (directly outputting the posterior mean).

    Input:
        x_t: noisy image at timestep t
        t: current timestep
        mean_theta: network-predicted posterior mean  $\hat{\mu}_{\theta}(x_t, t)$ 
    Output:
        sample_prev: denoised image sample at timestep t-1
    """
    ##### TODO #####

    sample_prev = None
    #####
    return sample_prev
```

Task2 – Image Generation

TODO #4 – compute loss

- image_diffusion_todo/model.py

1. Mean predictor $\mu_\theta(\mathbf{x}_t, t)$:

$$\mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)}[\|\mu_\theta(\mathbf{x}_t, t) - \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)\|^2]$$

2. $\mathbf{x}_0$ predictor $\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)$:

$$\mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)}[\|\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t) - \mathbf{x}_0\|^2]$$

3. ϵ_t predictor $\hat{\epsilon}_\theta(\mathbf{x}_t, t)$:

$$\mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)}[\|\hat{\epsilon}_\theta(\mathbf{x}_t, t) - \epsilon_t\|^2]$$

```
def get_loss(self, x0, class_label=None, noise=None):
    if self.predictor == "noise":
        return self.get_loss_noise(x0, class_label, noise)
    elif self.predictor == "x0": #### TODO
        return self.get_loss_x0(x0, class_label, noise)
    elif self.predictor == "mean": #### TODO
        return self.get_loss_mean(x0, class_label, noise)
    else:
        raise ValueError(f"Unknown predictor: {self.predictor}")
```

```
def get_loss_x0(self, x0, class_label=None, noise=None):
    ##### TODO #####
    # Here we implement the "predict x0" version.
    # 1. Sample a timestep and add noise to get (x_t, noise).
    # 2. Pass (x_t, timestep) into self.network, where the output should represent the clean sample x0_pred.
    # 3. Compute the loss as MSE(predicted x0_pred, ground-truth x0).
    #####
    loss = None
    return loss

def get_loss_mean(self, x0, class_label=None, noise=None):
    ##### TODO #####
    # Here we implement the "predict mean" version.
    # 1. Sample a timestep and add noise to get (x_t, noise).
    # 2. Pass (x_t, timestep) into self.network, where the output should represent the posterior mean mu_theta(x_t, t).
    # 3. Compute the *true* posterior mean from the closed-form DDPM formula (using x0, x_t, noise, and schedule).
    # 4. Compute the loss as MSE(predicted mean, true mean).
    #####
    loss = None
    return loss
```

Task2 – Image Generation

TODO #4.5 – verify (optional)

- `image_diffusion_todo/tests/test_todo.py`

Optional reference check: compare your implementation with the TA's. **Not graded.**

You can implement it your way, your score depends **only on the FID score and the report.**

This might prevent you from wasting your time for training with incorrect implementation.

Task2 – Image Generation

TODO #5 – train

- Run `image_diffusion_todo/dataset.py` to download the train & val dataset.
- Run `image_diffusion_todo/train.py` to train the diffusion model.
 - Please train for 50k – 100k iters for convergence.
 - It'll take around 6hrs or more per train, so start training ASAP.
- Run `image_diffusion_todo/sampling.py` on your trained ckpt to generate some images
- Run `image_diffusion_todo/fid/measure_fid.py` to evaluate your generated images' quality.

FID: 229.33834138594412

Incorrect FIDs

FID: 10.844529455220403

Correct FIDs

Task2 – Image Generation

TODO #5 – train

- You'll need to test with different beta schedulers and predictors
 - {linear, quadratic, cosine} scheduler + noise predictor
 - linear scheduler + {noise, x0, mean} predictor
- Observe the denoising process and result's quality.
 - Trajectory figures (step={step}-traj.png) show the denoising process



Task2 – Image Generation

TODO #6 – report

- Complete and Explain of all TODO codes (30pts).
- Show the trajectory figures and compare results across three beta schedulers (10pts)



- Show the results of different predictors and discuss. (10pts)



8 samples per predictor, 24 in total.

- Screenshot the best FID of your training result. (20pts)
 - 20 pts: $FID < 15$
 - 15 pts: $15 \leq FID < 20$
 - 10 pts: $20 \leq FID < 30$
 - 5 pts: $30 \leq FID < 40$
 - 0 pts: $FID \geq 40$

Submission

Create a single ZIP file {ID}_lab1.zip including

- The report PDF
- The code without training checkpoints, dataset, and the Inception Network weight.

See the README for detailed instruction

Due: 2026/10/1 (Thu.) 23:59. Please start early.